

Analyzing Social Media Usage

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Introduction

Social Media has become an integral part of modern life in terms of communication and keeping up with others. Understanding the impact that different social media platforms can have on users can help provide insight onto creating boundaries for ourselves when it comes to utilizing them. The purpose of this study is to analyze social media media usage with emotional well-being along with verifying if there are any relationships between daily usage among age categorized by gender.

Methodology

The analysis for this study was conducted using R, in particular utilizing the libraries dplyr for data manipulation and ggplot2 for visualizing the data. The study includes: 1. Identifying the most common emotions associated with each platform. 2. Ranking the platforms based on total time spent along with average daily time spent. 3. Analyzing the most common platforms for each gender. 4. Examining if there's a correlation between daily usage time and age, categorizing by different genders.

Data Set

The data set is sourced from Kaggle, licensed under Massachusetts Institute of Technology. It contains 10 variables (columns) along with 1001 entries (rows) before filtering. The variables include User ID, Age, Gender, Platform, Daily Usage Time (in minutes), Posts Per Day, Likes Received Per Day, Comments Received Per Day, Messages Sent Per Day, and Dominant Emotion. The variables included both numerical and categorical.

Which platform was the most common for each emotion?

```
platform_emotion <- social |>
  group_by(Dominant_Emotion) |>
  summarize(Most_Common_Platform = Platform[which.max((table(Platform)))])

print(platform_emotion)
```

```
## # A tibble: 6 x 2
##   Dominant_Emotion Most_Common_Platform
##   <chr>            <chr>
## 1 Anger            Twitter
## 2 Anxiety          Instagram
## 3 Boredom          LinkedIn
## 4 Happiness        Instagram
## 5 Neutral          Facebook
## 6 Sadness          Instagram
```

Results

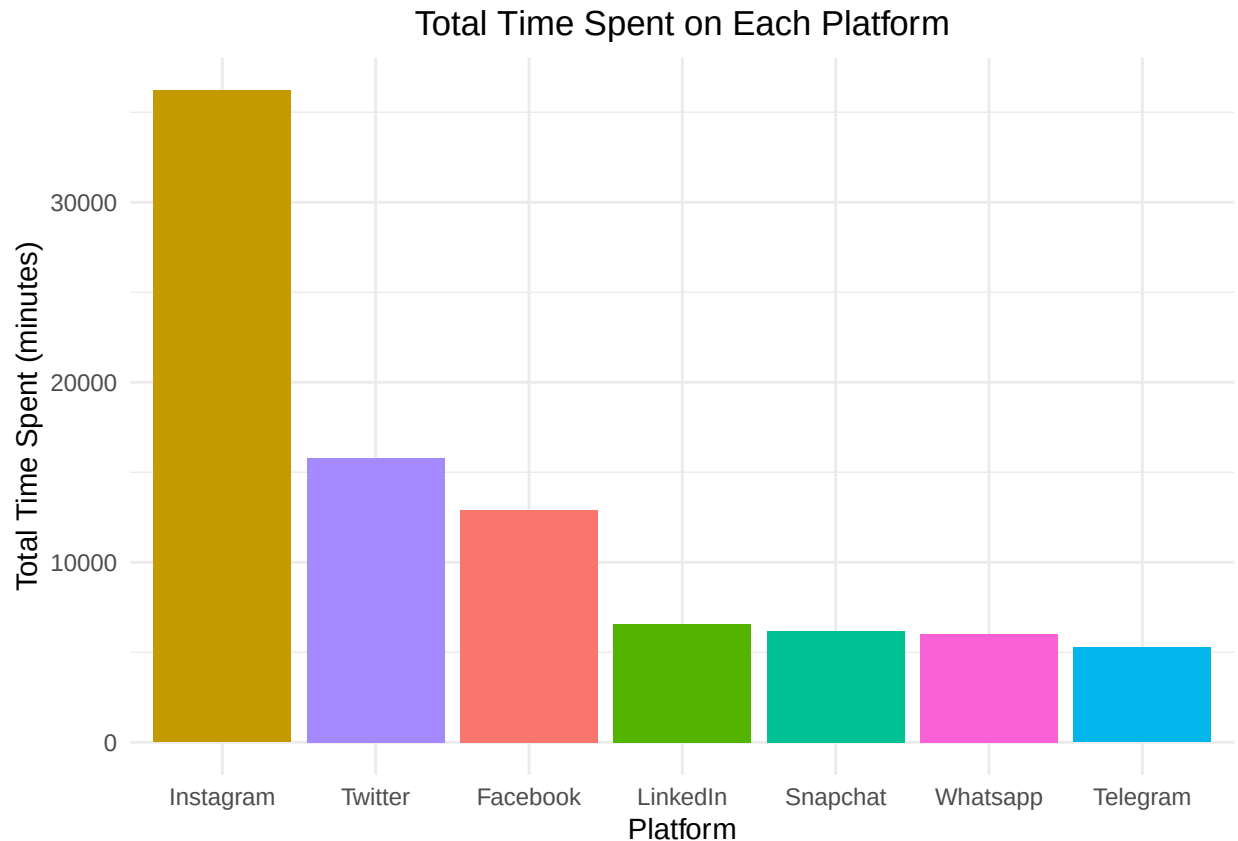
The results for most dominant associated with each platform seem to align with the general consensus. Twitter is typically a very political charged platform, LinkedIn is used for job-hunting, and Facebook is typically used as a functional platform for messaging, purchasing items, etc. The most interesting results come from Instagram, which were associated with a variety of emotions including Anxiety, Happiness, and Sadness. Ranking the platforms with the most time spent on it

```
time_spent_platform <- social |>
  group_by(Platform) |>
  summarise(Total_Time_Spent = sum(`Daily_Usage_Time..minutes.`, na.rm = TRUE)) |>
  arrange(desc(Total_Time_Spent))

print(time_spent_platform)
```

```
## # A tibble: 7 x 2
##   Platform Total_Time_Spent
##   <chr>         <dbl>
## 1 Instagram      36220
## 2 Twitter         15800
## 3 Facebook       12900
## 4 LinkedIn        6580
## 5 Snapchat        6190
## 6 Whatsapp        6030
## 7 Telegram       5270
```

```
ggplot(time_spent_platform, aes(x=reorder(Platform,-Total_Time_Spent), y=Total_Time_Spent, fill=Platform)) +
  geom_bar(stat="identity") +
  labs(title="Total Time Spent on Each Platform",
       x="Platform",
       y="Total Time Spent (minutes)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5), legend.position = "none")
```



Results

Instagram had by far the most time spent on it with 36,220 minutes. More than double the second most Platform (Twitter). In terms of understanding the results, it makes sense that Telegram, Whatsapp, and Snapchat have the least amount of time spent since they are primarily used for communication. Instagram, Twitter, and Facebook to some extent (The top 3) are also used for entertainment purposes.

Ranking the platforms with the most time spent on it

```
# Filter out rows with NA values in Daily Usage Time
social <- social |> filter(!is.na(Daily_Usage_Time..minutes.))

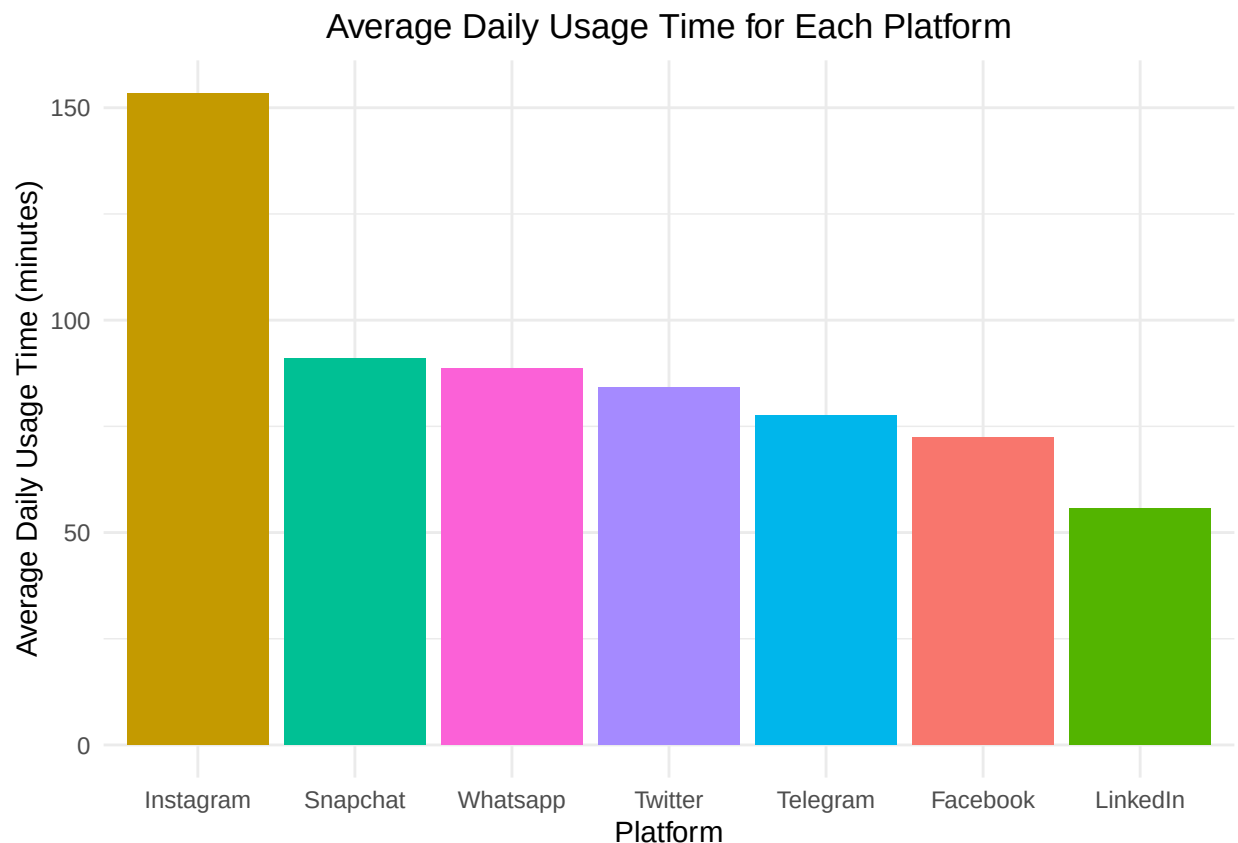
# Calculate the average daily usage time for each platform
average_time_spent_platform <- social |>
  group_by(Platform) |>
  summarise(Average_Daily_Usage_Time = mean(Daily_Usage_Time..minutes., na.rm = TRUE)) |>
  arrange(desc(Average_Daily_Usage_Time))

# Print the result
print(average_time_spent_platform)
```

```
## # A tibble: 7 x 2
```

```
## Platform Average_Daily_Usage_Time
## <chr> <dbl>
## 1 Instagram 153.
## 2 Snapchat 91.0
## 3 Whatsapp 88.7
## 4 Twitter 84.0
## 5 Telegram 77.5
## 6 Facebook 72.5
## 7 LinkedIn 55.8
```

```
# Create a bar plot for the average daily usage time for each platform
ggplot(average_time_spent_platform, aes(x=reorder(Platform, -Average_Daily_Usage_Time),
y=Average_Daily_Usage_Time,
fill=Platform)) +
  geom_bar(stat="identity") +
  labs(title="Average Daily Usage Time for Each Platform",
x="Platform",
y="Average Daily Usage Time (minutes)") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5), legend.position = "none")
```



Similar to most time spent on each platform, the average daily usage time for each platform yielded Instagram as the highest. What was interesting was that the rest of the results did not match up with the most time spent on each platform. The rest of the platforms were in a similar ballpark of ~75-90 minutes of average daily usage with the notable exception of LinkedIn which was closer to an average of 50 minutes per day.

Most common platform for each gender

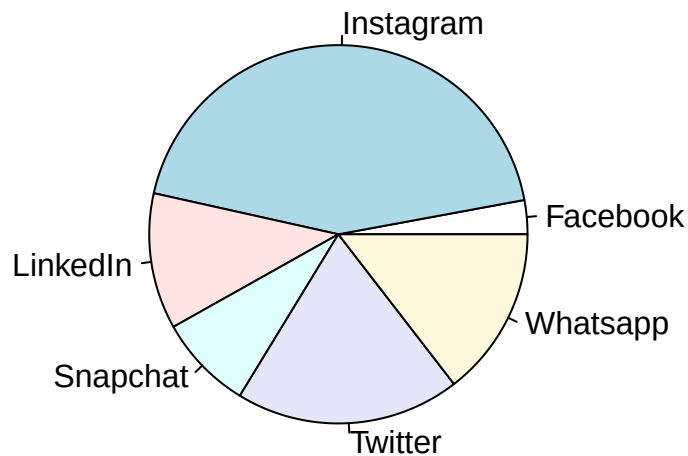
```
# Pie Chart of Each Platform per Gender:
filtered_social <- social |> filter(!(Gender %in% c('Marie', '27')))

unique_genders <- unique(filtered_social$Gender)

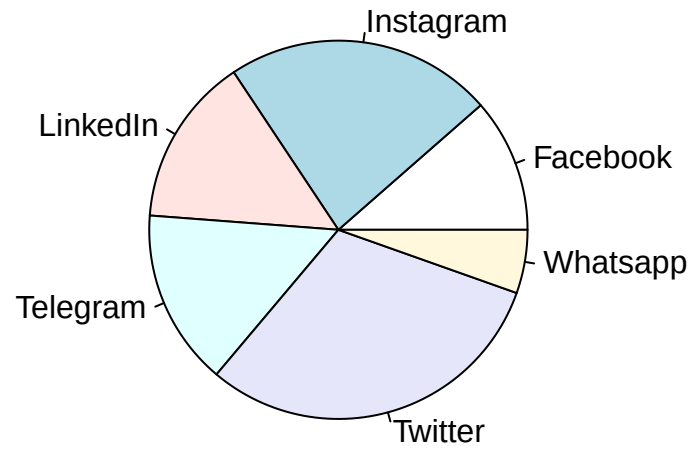
for (gender in unique_genders)
{
  gender_data <- filtered_social |> filter(Gender == gender)
  platform_counts <- table(gender_data$Platform)

  pie(platform_counts,
      main=paste("Platform Distribution for", gender))
}
```

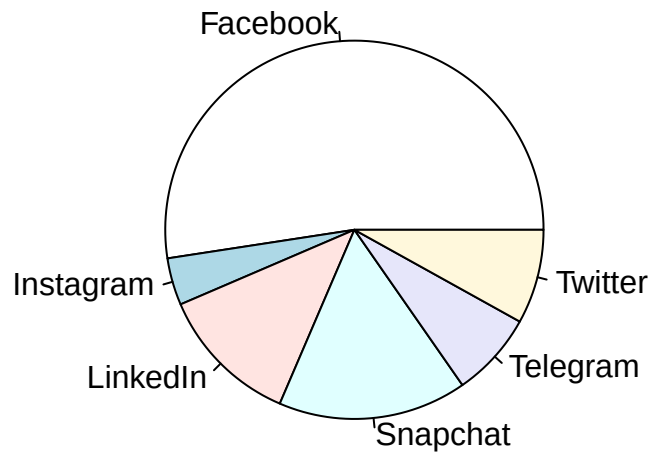
Platform Distribution for Female



Platform Distribution for Male



Platform Distribution for Non-binary



Categorizing the distribution of platform use by gender, the most dominant platform for Females was Instagram, for Males it was Twitter, and for Non-binary it was Facebook. Visually it looks like the Male distribution was the most spread out, and the least was Non-binary.

Analyzing the relationship between Daily Usage Time and Age for each gender

```
analysis <- function(gender_data, gender){  
  # Correlation coefficient  
  correlation <- cor(gender_data$Age,  
                    gender_data$Daily_Usage_Time..minutes., use = "complete.obs")  
  
  # linear regression  
  lm_model <- lm(Daily_Usage_Time..minutes. ~ Age, data=gender_data)  
  lm_summary <- summary(lm_model)  
  
  cat("Gender: ", gender, "\n")  
  print(lm_summary)  
  
  # Scatter plot with regression line:  
  p <- ggplot(gender_data, aes(x=Age, y=Daily_Usage_Time..minutes.)) +  
    geom_point() +  
    geom_smooth(method="lm", se=TRUE, color="red") +
```

```

    labs(title=paste('Daily Usage Time vs Age for ', gender),
          x='Age',
          y='Daily Usage Time (Minutes)') +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.5))

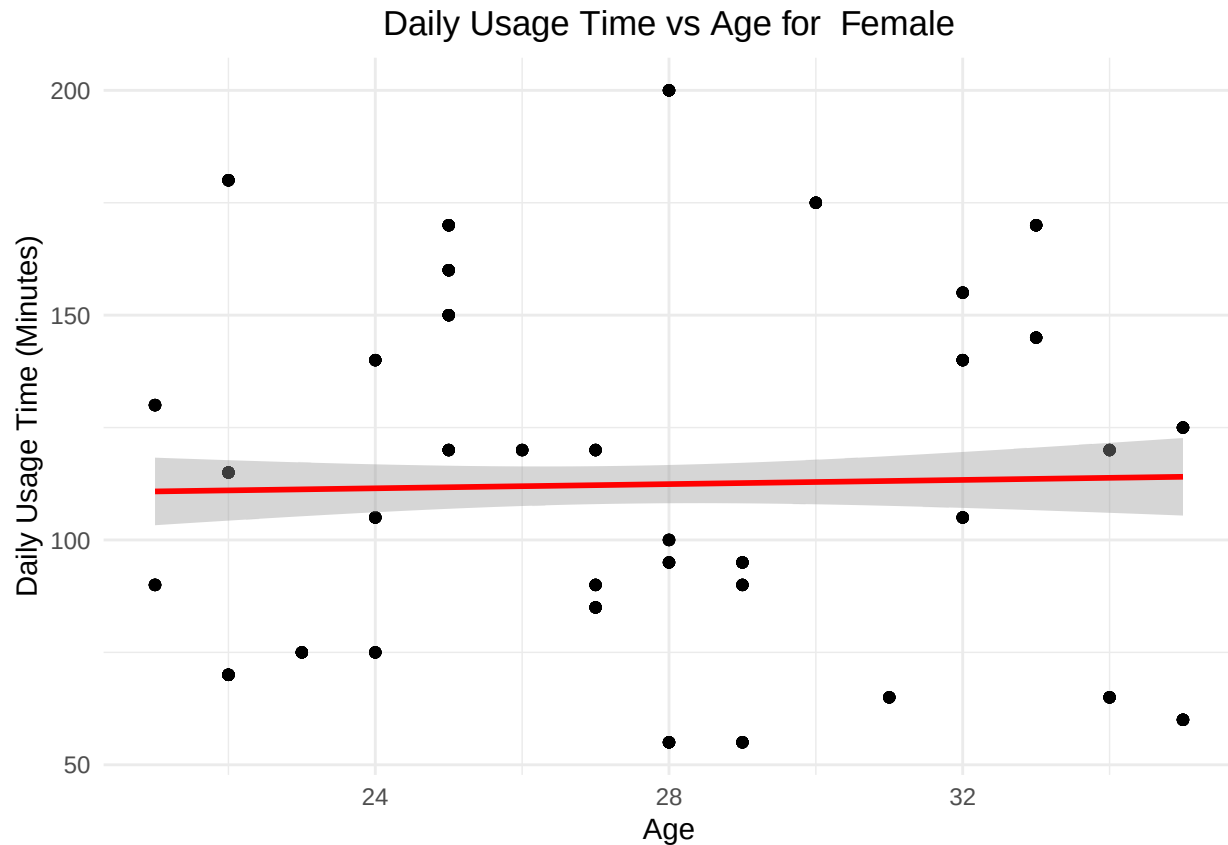
  print(p)
}

# Perform analysis for each gender:
for (gender in unique_genders)
{
  gender_data <- social |> filter(Gender == gender)
  analysis(gender_data, gender)
}

## Gender: Female
##
## Call:
## lm(formula = Daily_Usage_Time..minutes. ~ Age, data = gender_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -57.66 -36.26  -6.49   28.51   87.58
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 105.9028    13.8231   7.661 1.91e-13 ***
## Age          0.2328     0.4996   0.466  0.641
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 39.65 on 342 degrees of freedom
## Multiple R-squared:  0.0006347, Adjusted R-squared:  -0.002287
## F-statistic: 0.2172 on 1 and 342 DF, p-value: 0.6415

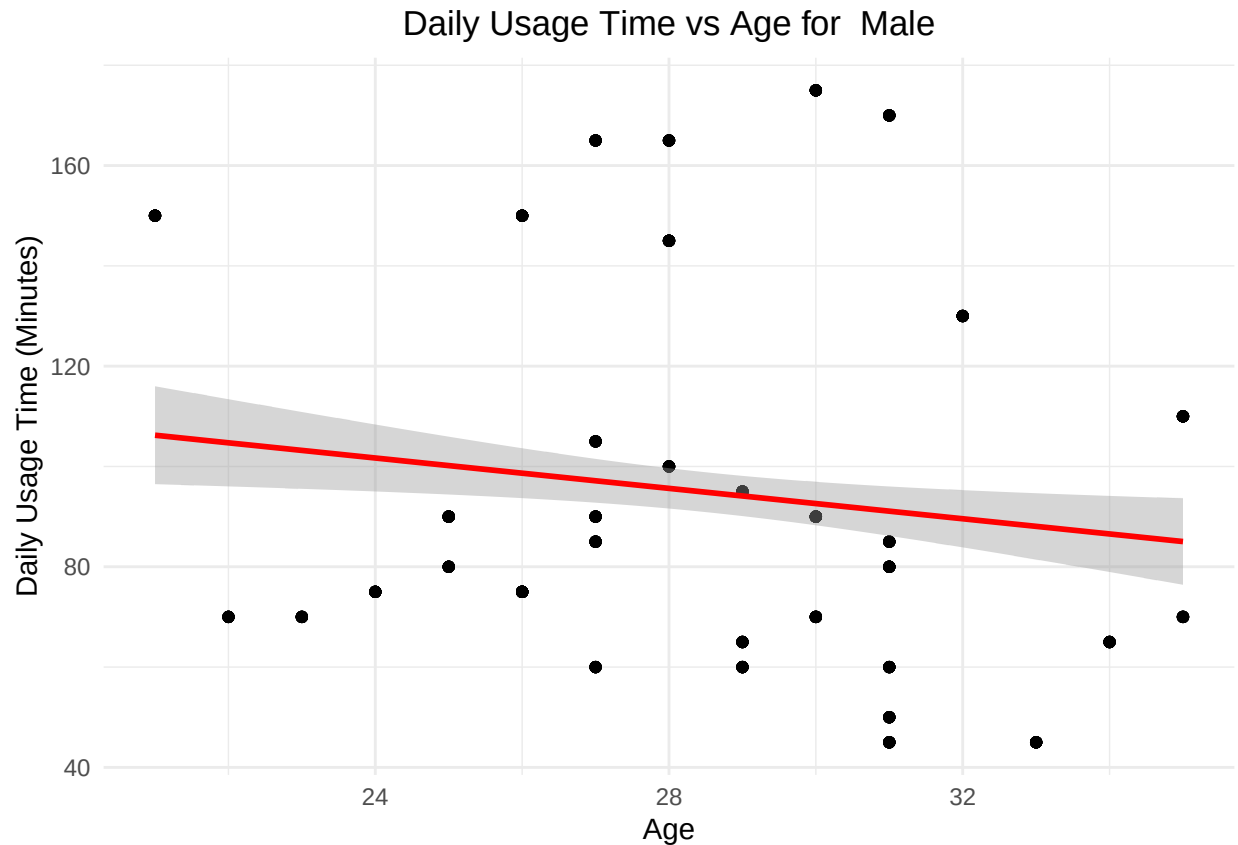
## 'geom_smooth()' using formula = 'y ~ x'

```

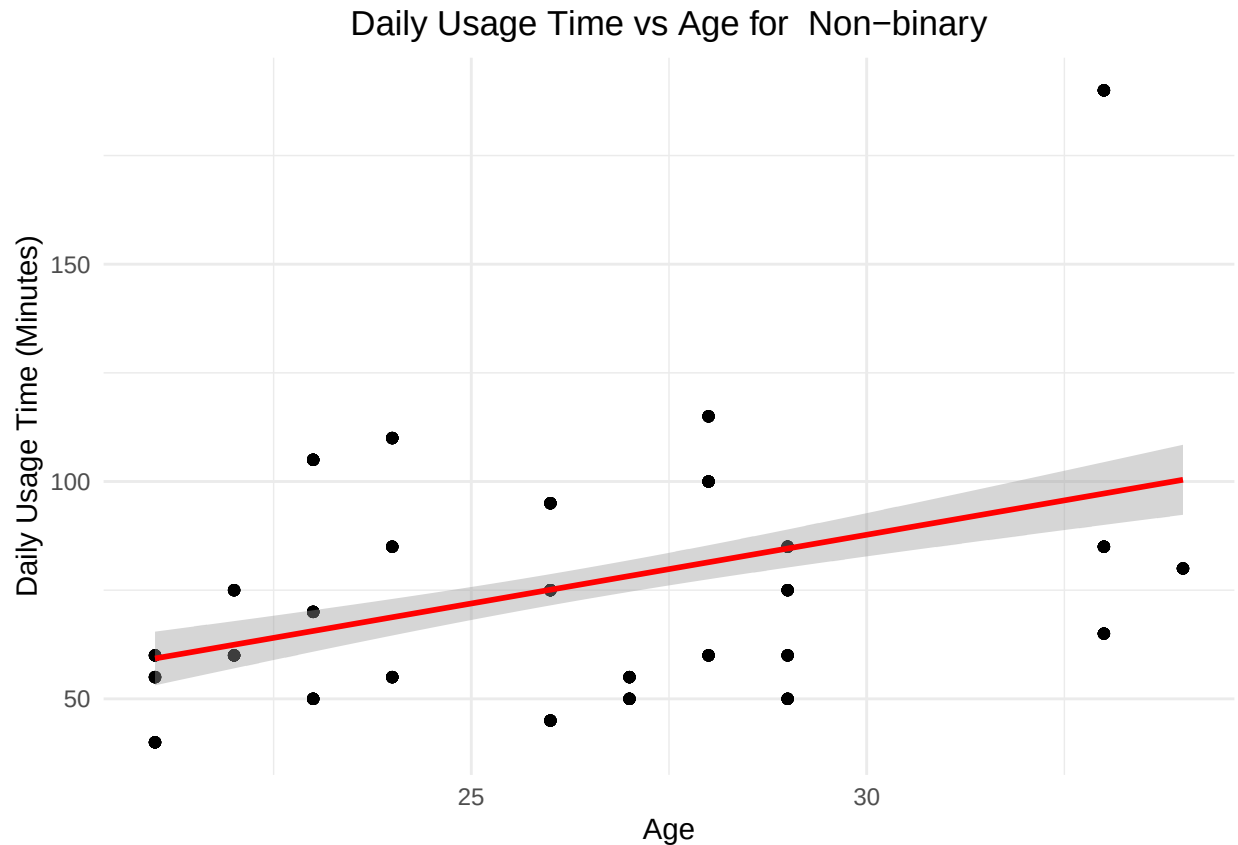
```
## Gender:  Male
##
## Call:
## lm(formula = Daily_Usage_Time..minutes. ~ Age, data = gender_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -46.09 -26.69 -10.17  24.97  82.40
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 138.0351    17.2783   7.989 2.3e-14 ***
## Age         -1.5144     0.6013  -2.518 0.0123 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 36.79 on 330 degrees of freedom
## Multiple R-squared:  0.01886,    Adjusted R-squared:  0.01588
## F-statistic: 6.342 on 1 and 330 DF,  p-value: 0.01226

## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Gender:  Non-binary
##
## Call:
## lm(formula = Daily_Usage_Time..minutes. ~ Age, data = gender_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -34.586 -21.424  -4.284  16.227  92.763
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -7.1331    12.7223  -0.561   0.576
## Age           3.1627     0.4781   6.616 2.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 28.67 on 246 degrees of freedom
## Multiple R-squared:  0.151, Adjusted R-squared:  0.1476
## F-statistic: 43.77 on 1 and 246 DF, p-value: 2.29e-10

## 'geom_smooth()' using formula = 'y ~ x'
```



```
# Female equation: Time_Spent = 0.0027*Age + 105.9
#                   P-value = 0.2328
#                   R-squared = 0.0006347
# Male equation: Time_Spent = -1.5144*Age + 138.0
#                   P-value = 0.0123
#                   R-squared = 0.01886
# Non-Binary equation: Time_Spent = 3.1627*Age - 7.13
#                   P-value = 0.2.29e-10
#                   R-squared = 0.151
```

Results

For females, the P-value > 0.05 indicating that it is not statistically significant. For males, there is some statistical significance since the P-value is less than 0.05. This means there is a slight decrease in daily usage of social media with age. For Non-Binary individuals the P-value was $2.29e-10$, indicating a high statistical significance between social media usage and age with the R value indicating that it explains 15.1% of the variance. In conclusion, for females, age is not a contributing factor for usage of social media. For men, there's a slight decrease with age and for Non-binary, there is a significant increase in social media usage among older individuals.

Conclusion

In conclusion, this study showcased some of the emotions, time spent, and revealed some trends associated with Age and time spent among different social media platforms. Instagram was the most commonly used social media with the most positive and negative emotions associated with it. Age was a contributing factor to time spent using social media for Non-binary individuals, and to some extent males as well. In terms of future studies, it would be interesting to look at a wider range of emotions among individuals and to include other social media apps such as TikTok, MeetUp, and dating apps. In terms of what I've learned, this project made me realize the difficulties in gathering data. Kaggle made it easier, but sourcing through government websites is difficult and it is hard to find detailed data sets. This made me realize that much of the major battles in data analysis is compiling the data together properly and filtering.